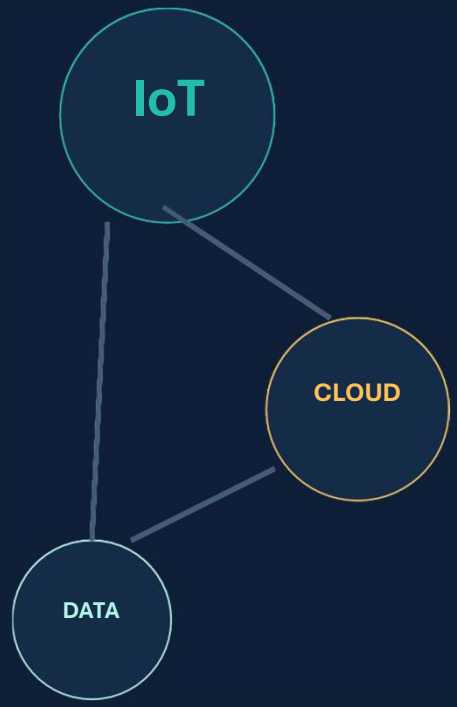

INTERNET OF THINGS
& CLOUD COMPUTING
GPS Map Camera
Mevalurkuppam, Tamil Nadu, India
Ece Department, Mevalurkuppam, Tamil Nadu 602105
Lat 13.026011° Long 80.016139°
Thursday, 15/10/2026 10:35 AM GMT +05:30
Note : Captured by GPS Map Camera
Google

TECHNICAL PRESENTATION | CSA15

INTERNET OF THINGS G CLOUD COMPUTING

Building Connected Ecosystems

How connected devices collect, process and share data through cloud platforms.



Presented by: Muhitha K
Register Number: 192524339

What are IoT and Cloud Computing?

Two technologies that work together to make connected services useful.

- **Internet of Things (IoT):** physical devices with sensors/software that exchange data over a network.
- **Cloud computing:** on-demand computing, storage and services delivered over the internet.
- **Why combine them?** IoT devices generate data; cloud platforms store and process it at scale.
- **Remote visibility:** users can view device status and readings from connected applications.
- **Automation:** rules or analytics can trigger actions when conditions are met.
- **Presentation objective:** understand the architecture, benefits, challenges and a practical use case.

How an IoT System Works

An IoT solution turns physical events into data that software can use.

1

Sensors G devices

Measure conditions such as temperature, motion, light or soil moisture.

2

Connectivity

Wi-Fi, cellular, Bluetooth or other networks carry readings.

3

Data collection

A device or gateway gathers and forwards the measurements.

4

Processing

Software filters, analyzes or compares incoming data with rules.

5

User interface

A dashboard or mobile app displays status, alerts and trends.

6

Actuation

Where supported, a command can switch equipment on or off.

IoT–Cloud Architecture G Data Flow

01

SENSORS / DEVICES

Capture real-world measurements

02

GATEWAY / NETWORK

Transmit data securely through the internet

03

CLOUD PLATFORM

Store and process data at scale

04

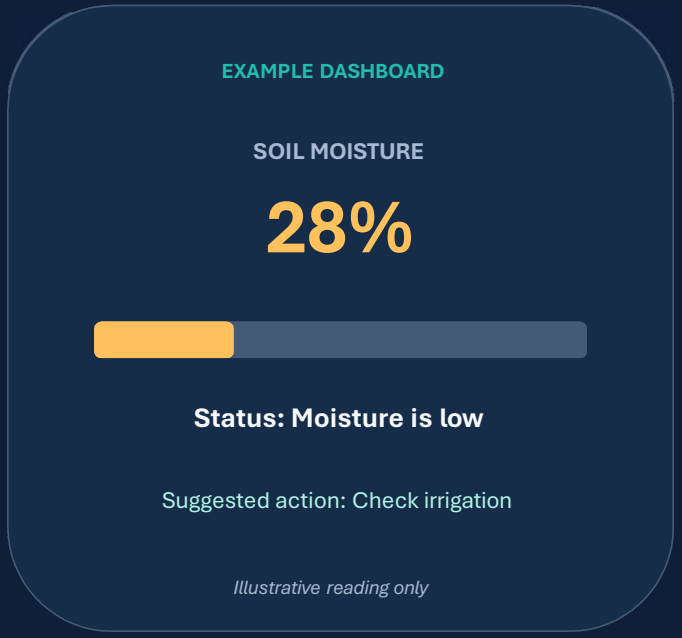
APP / DASHBOARD

Show readings, trends and alerts

Smart Agriculture: From Sensor to Action

A simple example of IoT and cloud computing solving a real-world problem.

- **Measure:** a soil-moisture sensor checks how wet or dry the soil is.
- **Send:** a connected device or gateway transmits the reading.
- **Store:** the cloud keeps readings over time for comparison.
- **Analyze:** a rule checks whether moisture is below a chosen threshold.
- **Alert or act:** the farmer receives an alert, or compatible irrigation is activated.
- **Improve:** historical data helps plan watering and reduce unnecessary use of water.



Applications, Benefits G Challenges

The same IoT–cloud pattern appears across many sectors.

INDUSTRY USES

- Healthcare: wearable monitoring
- Smart homes: lights and thermostats
- Agriculture: soil and irrigation sensors
- Cities: traffic and air-quality monitoring
- Industry: machine-condition monitoring

KEY BENEFITS

- Remote monitoring and control
- Centralized data storage
- Scalable computing resources
- Faster alerts and decisions
- Historical data for planning

CHALLENGES

- Privacy and data protection
- Weak or unreliable connectivity
- Device security and updates
- Latency for time-critical actions
- Cost, compatibility and maintenance

Good design balances convenience with security, reliability and responsible data handling.

Conclusion G Key Takeaway

IoT and cloud computing create value when devices, data and applications work together.

- **IoT connects:** sensors and devices capture information from the physical world.
- **The cloud enables:** centralized storage, scalable processing and analytics.
- **Together they support:** monitoring, alerts, automation and better decisions.
- **Real-world value:** smart agriculture can help farmers respond to soil conditions.
- **Main challenge:** security, privacy and dependable connectivity must be planned.
- **Key takeaway:** IoT collects the data; cloud computing helps turn it into useful action.